FROM POLICY TO PRODUCT: RWANDA'S JOURNEY IN CROP BIOTECHNOLOGY, CLIMATE RESILIENCE, AND AGRICULTURAL COMPETITIVENESS

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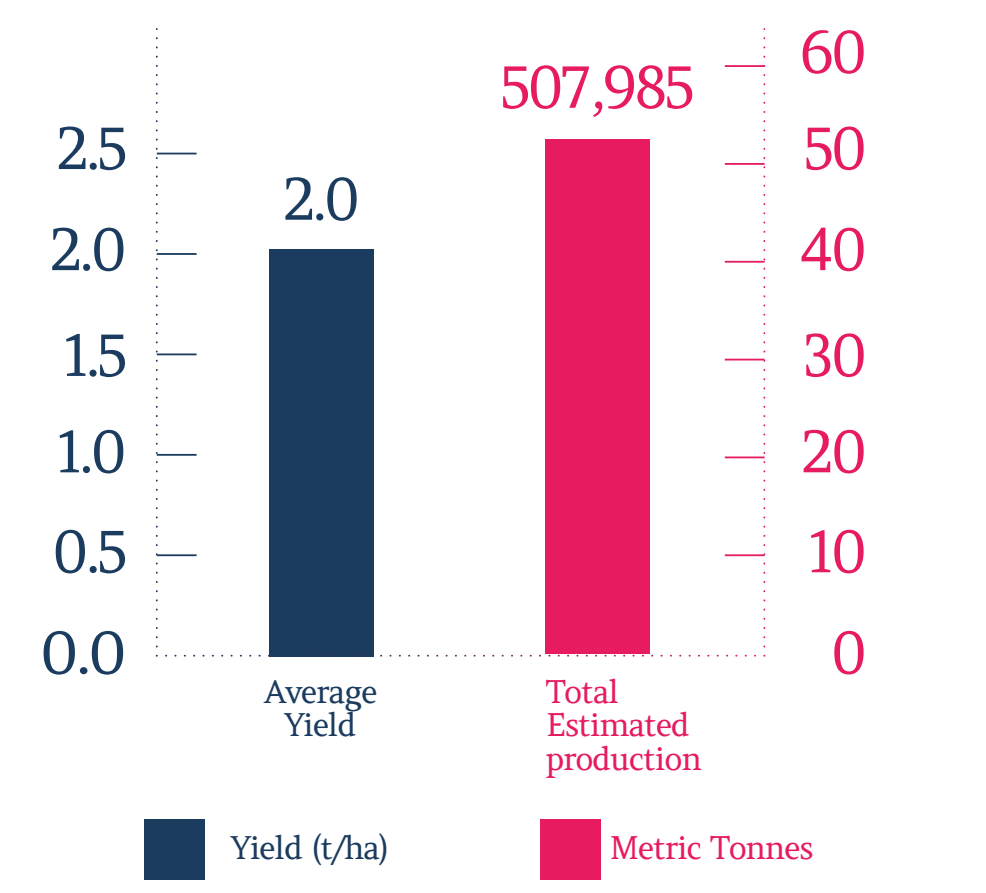
INTRODUCTION

Rwanda's agriculture is shaped by steep hills, altitude differences, rainfall variability, and diverse agroecological zones.[1] This supports maize, Irish potato, cassava, and banana production, but it also increases exposure to pests, diseases, and climate stress.[1 - 2]

Cassava, banana, and Irish potato face major threats: Fall Armyworm and Drought in maize; Late Blight in potato; CMD and CBSD in cassava; and BXW and Fusarium wilt risks in banana.[2 - 5]



In Season A 2024, maize averaged about 2.0 t/ha, with production estimated at 507,985 metric tons.[2]

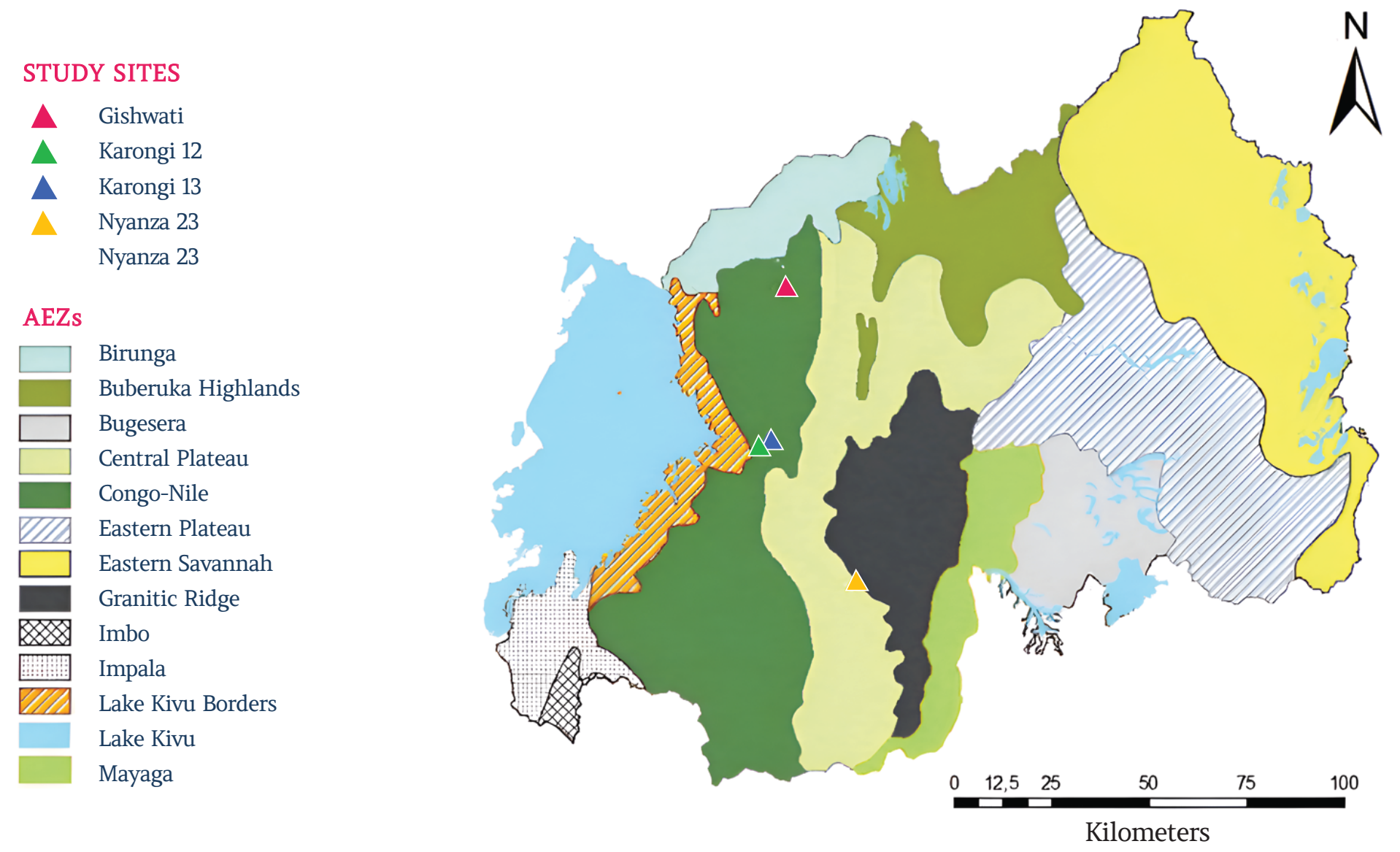
Rwanda's investments in innovation, biosafety governance, digital transformation, and modern agricultural research capacity[3 - 5] have supported a shift from policy preparation to responsible biotechnology product development.

Rwanda signed the Cartagena Protocol on Biosafety in 2000 and ratified it in 2004[4] provideing the national framework for permits, risk assessment, decision-making, monitoring, and safe use of living modified organisms.[4]

Regional and international partnerships

Rwanda is positioning crop biotechnology as a strategic tool for climate-smart seed systems, agricultural competitiveness, private sector participation, and improved smallholder farmers' livelihoods.1 - 5

Map of Rwanda showing its agro-ecological zones and study sites



WHY BIOTECHNOLOGY MATTERS

Biotechnology matters because it can help Rwanda develop better seed faster. It complements conventional breeding. It targets traits that farmers need most: pest resistance, disease resistance, drought tolerance, lower production costs, and higher yield stability.[2,6-8]

STUDIES FROM OTHER COUNTRIES

NIGERIA (TELA Maize)

↑ 88%
yield advantage compared to non-TELA maize

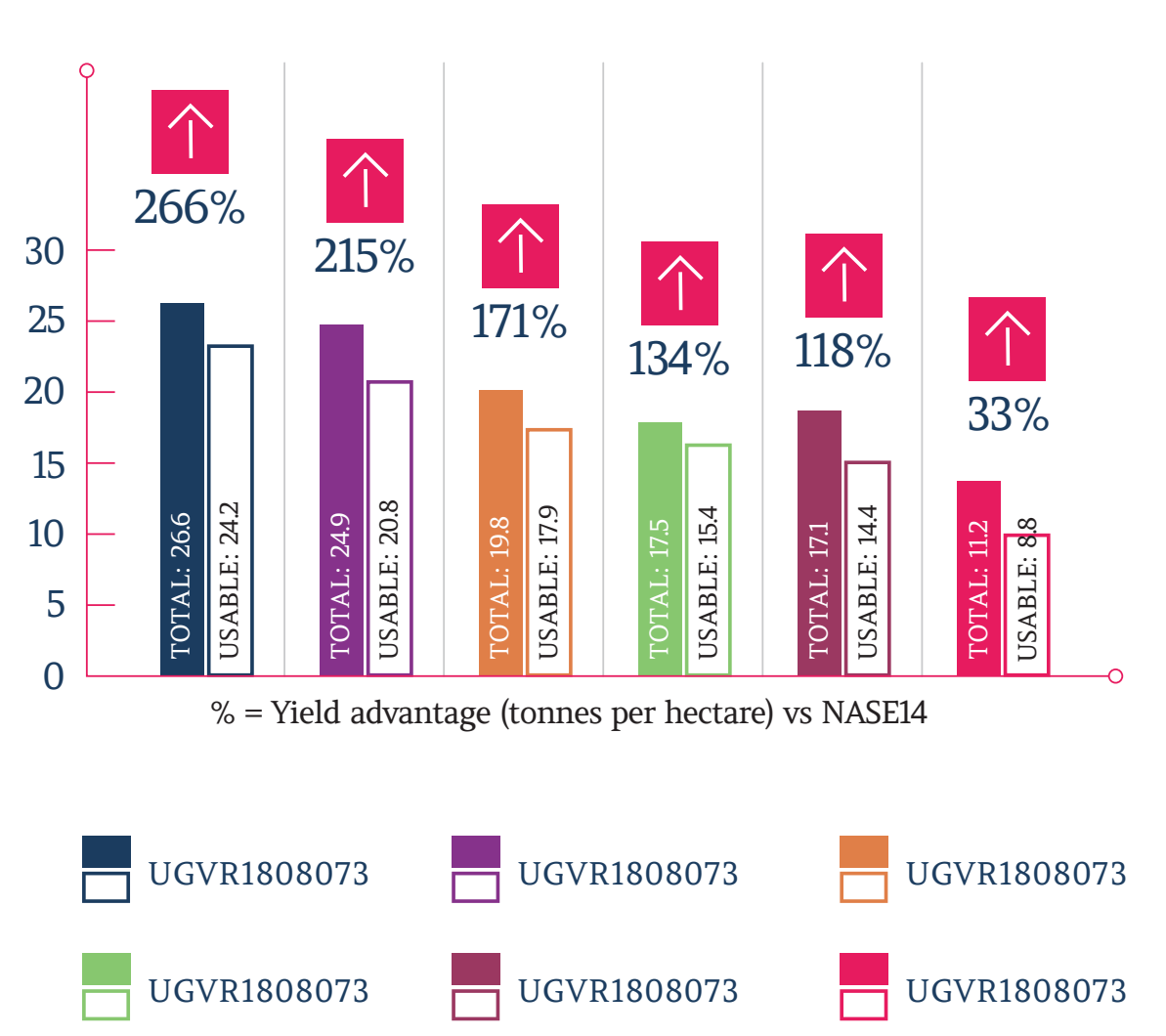
BANGLADESH (BT eggplants)

↑ 51%
yield increase

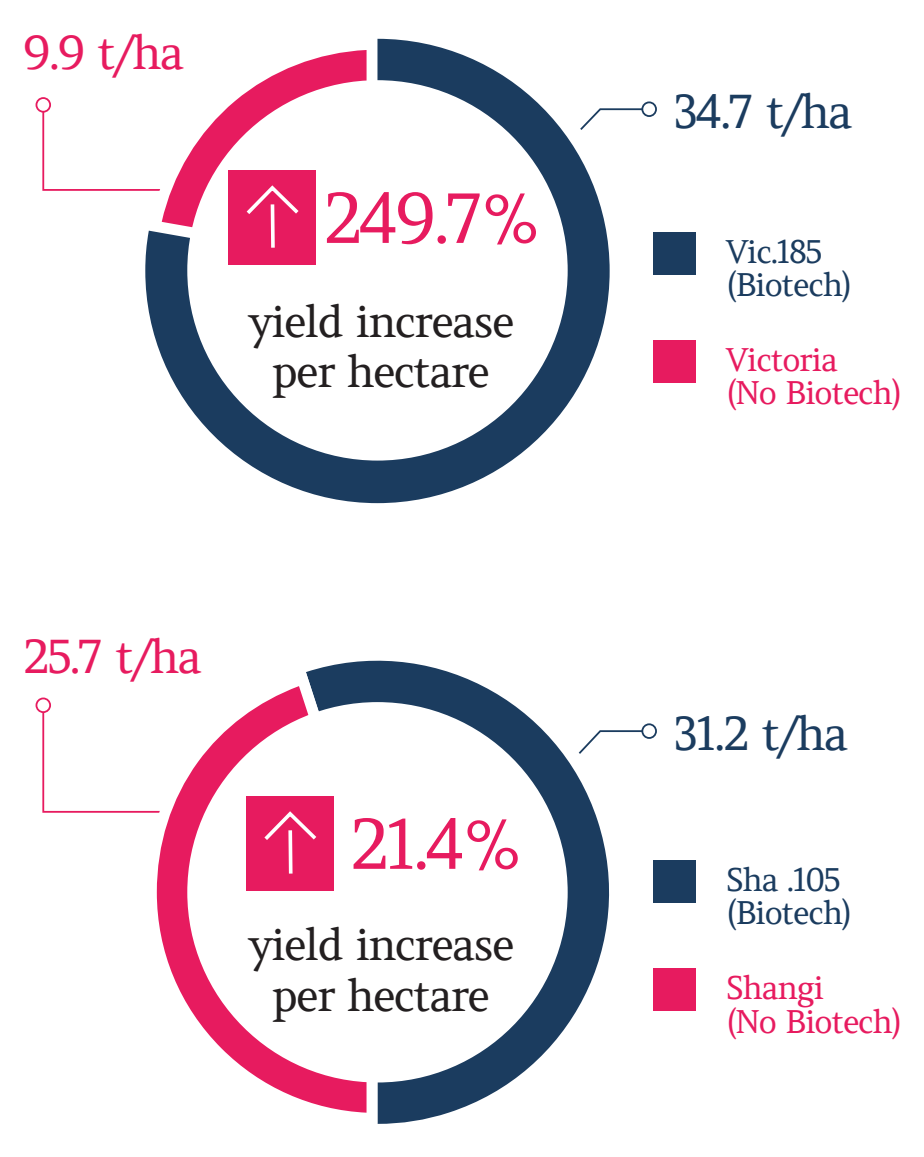
↑ 37.5%
reduction in pesticide costs

↑ 128%
increase in net revenue

Clone Perfomance and Yield Advantage

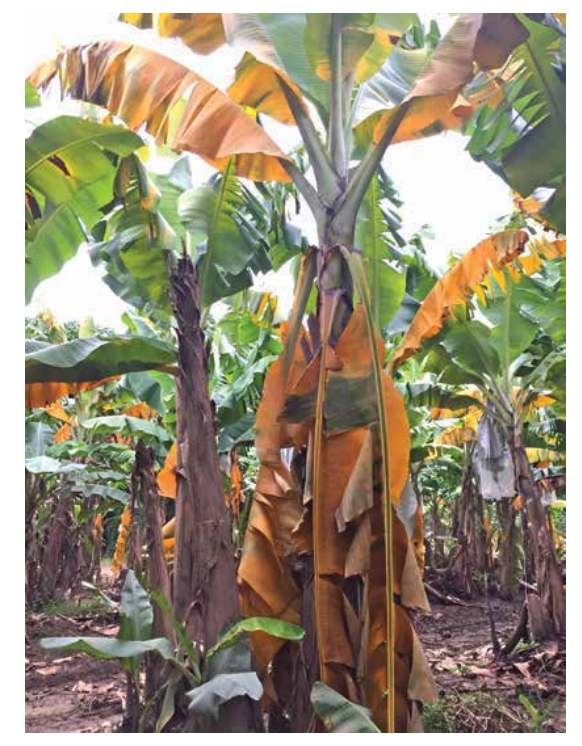


POTATO YIELD DATA: Biotech vs No Biotech



AGRICULTURAL CHALLENGE FACING FARMERS

BANANA CHALLENGE



Banana Bacterial Wilt

Fusarium Wilt Tropical Race 4

TR4 is especially concerning because it is soil-borne, persistent, and widely recognized as one of the most serious global threats to banana production.

THE SOLUTION

Development of locally preferred banana varieties with resistance to BXW and TR4, using genetic engineering and gene-editing approaches. This supports long-term productivity, disease resilience, and protection of farmer-preferred banana types.

CASSAVA CHALLENGE

Cassava Mosaic Disease (CMD)

Cassava Brown Streak Disease (CBSD)

CBSD is especially damaging because symptoms may be hidden in roots until harvest, and can cause losses of "up to 100% in susceptible varieties."

THE SOLUTION

Development of virus-resistant cassava targeting CMD and CBSD to stabilize yields, improve root quality, reduce losses, and support household food security and processing opportunities.

IRISH POTATO CHALLENGE

Potatos harvest with Late Blight disease

Farmers often rely on repeated fungicide applications, which increase production costs, labor demand, environmental exposure, and farmer health risks.

THE SOLUTION

Development of locally preferred late blight-resistant potato varieties to reduce fungicide dependence, lower production costs, improve yields, and protect both farmer health and the environment.

MAIZE CHALLENGE

Fall Armyworm

Stem borers

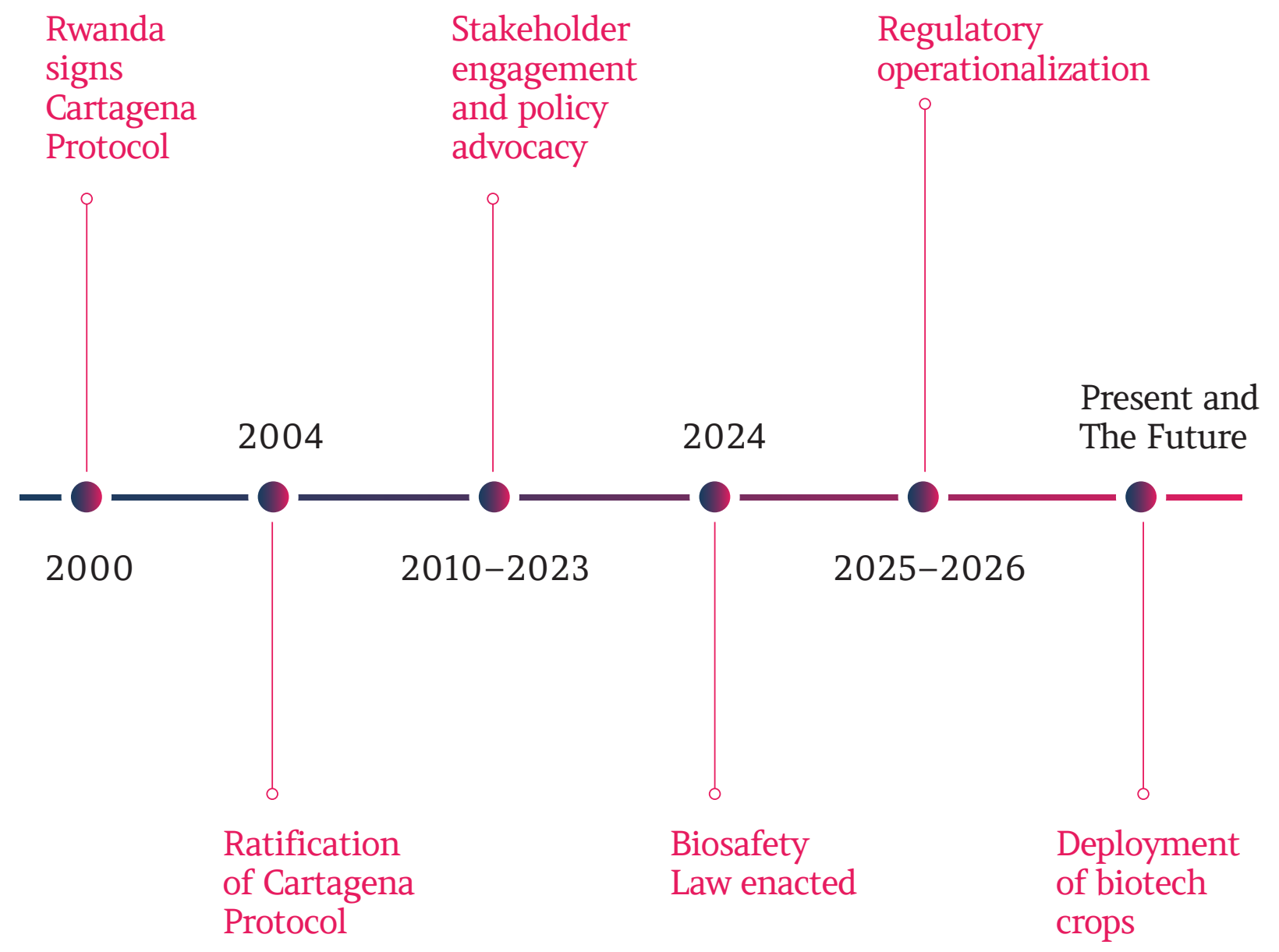
Rwanda's biotech maize communication materials describe biotech maize as able to protect against fall armyworm and stem borers, as well as moderate drought conditions.

THE SOLUTION

Development of maize varieties with resistance to fall armyworm and stem borers, combined with drought tolerance traits, to reduce crop losses, pesticide costs, and climate-related yield instability.

RWANDA'S POLICY AND BIOSAFETY JOURNEY

Rwanda's biosafety journey shows steady progress from international commitment to national implementation. The country signed the Cartagena Protocol in 2000, ratified it in 2004, and enacted Law No. 025/2024 to guide biosafety regulation.[4] The result is a clearer route from research to field evaluation, environmental release, variety registration, commercialization, and farmer adoption.[4, 6 - 8]



2000 – Rwanda signs Cartagena Protocol
2004 – Ratification of Cartagena Protocol
2010–2023 – Stakeholder engagement and policy advocacy
2024 – Biosafety Law enacted
2025–2026 – Regulatory operationalization
Present and The Future – Deployment of biotech crops

THE POWER OF COMMUNICATION AND STAKEHOLDER ENGAGEMENT

EDUTAINMENT (RADIO DRAMA SERIES)

SOCIAL MEDIA & DIGITAL INFLUENCERS

MEDIA ENGAGEMENT & MODERATION

EXPERIENTIAL COMMUNICATION

Rwanda's crop biotechnology deployment relies on public trust, stakeholder engagement, and evidence-based dialogue.

Initiatives like OFAB, RAB, and the BioCap Project foster science communication through radio drama, digital storytelling, social media influencers, and agripreneurs to translate complex crop biotechnology science into farmer-centered messages

FROM RESEARCH TO PRODUCT

Rwanda's biotechnology pipeline focuses on four food security crops: Banana, Cassava, Potato, and Maize. The goal is simple: protect yield, reduce losses, lower input pressure, and deliver farmer-preferred varieties that perform under local conditions.[6 - 8] The pipeline also uses regional evidence and data transportability where appropriate.[4, 6 - 8]

BBW & TR4 Resistance

Under Rwanda BioCap Project, the objective is to use the human resource capacity and infrastructure being built to develop and deploy two farmer-preferred banana varies (Injagi and Kamara) using gene editing technology.

CMD & CBSD Resistance

The Rwanda Agri-Biotech Program is testing six new virus-resistant cassava lines. Concurrently, the BioCap Project trains local scientists and upgrades infrastructure to enhance farmer-preferred crop varieties.

Late Blight Resistance

Under the RABP-CMP, Rwanda has completed the improvement of two locally grown varieties, Victoria and Ndamira/Shangi for late blight resistance while the Rwanda BioCap Project is building capacity to transform a locally farmer preferred Kingi Potato Variety for the late bright resistance.

Fall Armyworm & Drought Resistance

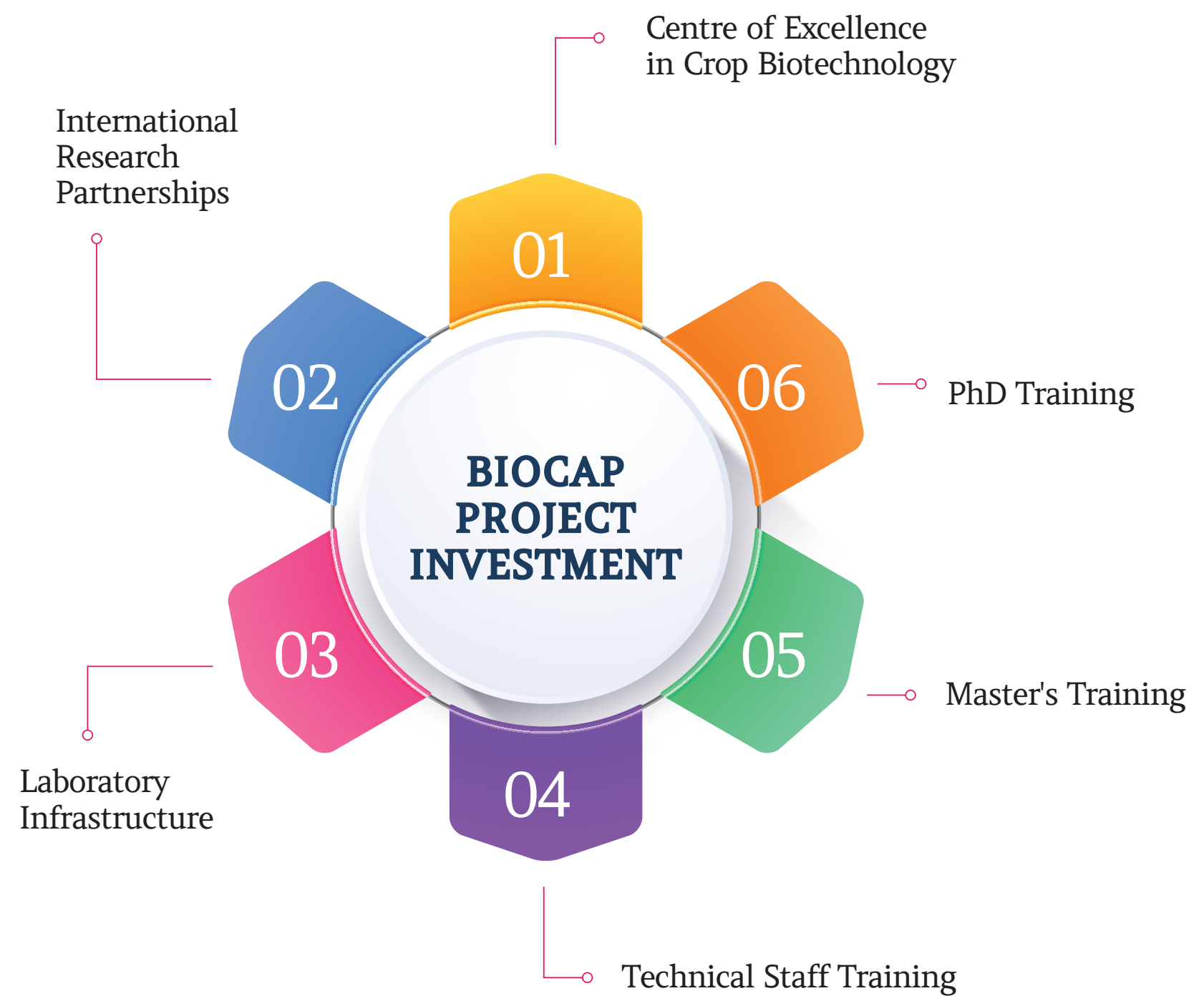
Among the top food-security crops, maize in Rwanda is being improved with BT gene technology using the TELA MAIZE technology which has already developed 12 GM maize varieties under efficacy and national performance trials in their advanced stage to variety registration.

BUILDING NATIONAL CAPACITY

The BioCap Project is building a Centre of Excellence in Crop Biotechnology at RAB Rubona.

It supports laboratories, screenhouses, tissue culture, molecular diagnostics, gene editing, bioinformatics, biosafety systems, and advanced staff training.7 - 8

Together, these investments move Rwanda from technology adoption toward technology development. The Centre of Excellence can become a national platform and regional hub for crop biotechnology research, training, biosafety, and product deployment.7 - 8



BIOCAP PROJECT INVESTMENT

01 Centre of Excellence in Crop Biotechnology
02 International Research Partnerships
03 Laboratory Infrastructure
04 Technical Staff Training
05 Master's Training
06 PhD Training

REGULATION, TRUST & TRANSPARENCY

Rwanda's biosafety system is built on science, transparency, public participation, and institutional accountability. This structure supports clear decisions from contained use and confined field trials to environmental release, market placement, import, export, transport, storage, and handling of living modified organisms.[4] Media training, stakeholder dialogue, field demonstrations, and farmer engagement help move biotechnology from technical debate to informed public understanding.[5 - 8]

REGULATORY INSTITUTIONS

National Biosafety Committee

PRIVATE SECTOR PARTICIPATION

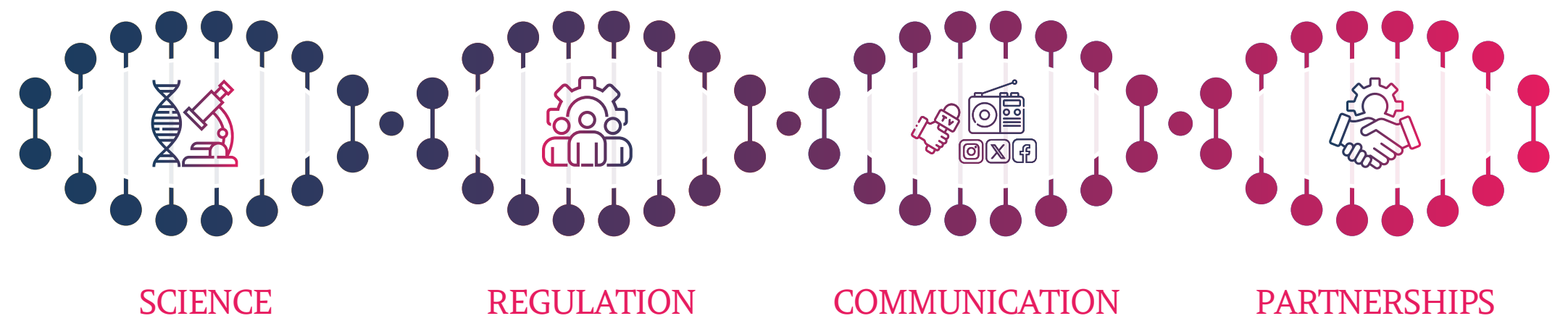
Rwanda has built a strong foundation through seed systems, public-private partnerships, and programmes that can multiply, commercialize, and deliver improved seed technologies to farmers.

Seed companies, seed multipliers, agro-dealers, processors, and agribusiness investors can help move biotechnology from trials to farms. Disease-resistant and climate-resilient varieties create opportunities for seed multiplication, youth agribusiness, and stronger rural enterprises.

RWANDA'S MODEL FOR BIOTECHNOLOGY SUCCESS

Rwanda is emerging as an example of how countries can responsibly use modern biotechnology to strengthen food security, climate resilience, and agricultural competitiveness. The country integrates science, regulation, communication, and partnerships into one coordinated biotechnology ecosystem. Rwanda's approach also uses regional collaboration and data transportability. This helps avoid unnecessary duplication while directing resources toward laboratories, advanced equipment, biosafety systems, and the training of local scientists.

FOUR PILLARS: TRUST & ADOPTION



SCIENCE

REGULATION

COMMUNICATION

PARTNERSHIPS

IMPACT AND FUTURE VISION

The expected impact is stronger food security, higher yield stability, lower crop losses, reduced pesticide pressure, improved farmer income, and national capacity to develop climate-smart seed.[2, 4, 6 - 12]
Through BioCap, Rwanda is investing in the practical application of biotechnology.

The project is building national capacity at RAB Rubona through a Centre of Excellence for Crop Biotechnology, advanced laboratories, staff training, biosafety support, and research on priority crops such as potato, cassava, and banana [14]. This investment moves Rwanda from policy commitment toward local product development and responsible deployment.

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1 World Bank Climate Change Knowledge Portal. Rwanda climate data and country-specific climate resources, including historical climatology and climate risk information.

2 National Institute of Statistics of Rwanda. Seasonal Agricultural Survey reports for 2023 and 2024, including crop yield, cultivated area, and production estimates for major crops.

3 World Bank Open Knowledge Repository. Building Science, Technology, and Innovation Capacity in Rwanda: Developing Practical Solutions to Practical Problems.

4 Government of Rwanda. Law No. 025/2024 of 16/02/2024 governing biosafety, Official Gazette Special Issue of 21 February 2024, and Cartagena Protocol-related biosafety commitments.

5 AUDA-NEPAD African Biosafety Network of Expertise, Rwanda Environment Management Authority, and partners. Biosafety capacity-building and regulatory strengthening activities in Rwanda.

6 Michigan State University Undergraduate Research. Poster presentation guidance and poster samples, including recommendations for poster size, short pitch preparation, and sample layouts.

7 Rwanda Agriculture and Animal Resources Development Board and AATF. Rwanda Agricultural Biotechnology Programme launch and annual review updates, including crop targets, partners, and programme milestones.

8 Food and Agriculture Organization and World Bank data platforms. Rwanda agriculture, food security, climate, and crop production context.

9 African Agricultural Technology Foundation. First Farmer Harvest of TELA Maize in Nigeria: Enhancing Farm-Level Yield and Revenues, 2025.

10 African Agricultural Technology Foundation. TELA Maize Enhancing Farm-Level Yield and Revenues: Case Study of Babban Gona Farmers, Nigeria, 2025.

11 Ahmed, A., Hoddinott, J. F., Abedin, N., and Hossain, N. The impacts of GM foods: Results from a randomized controlled trial of Bt eggplant in Bangladesh. American Journal of Agricultural Economics, 2021.

12 Hossain, M. A., Hossain, K. K., and Joshi, N. P. Impact of genetically modified Brinjal (Bt brinjal) on farmers' income and production in Pabna District, Bangladesh. GM Crops & Food, 2025.

13 Ministry of Agriculture and Animal Resources. Fifth Strategic Plan for Agriculture Transformation (PSTA5), 2024/25–2028/29.

14 Rwanda Agriculture and Animal Resources Development Board, CIP, IITA, and partners. Rwanda BioCap Project and Centre of Excellence for Crop Biotechnology launch information.